



Technical- Writing- Reviewer

Education (University of Caloocan City)

TECHNICAL WRITING

REVIEWER

MODULE 1

Technical writing is an audience-centered means of communication that provides a reader with clear and easy access to information. In the business world, time equates to profit, and profit is the force behind all business interaction. The technical writer and reader have a vis-à-vis relationship. The writer recognizes, respects, and addresses the importance of time in effective and efficient communication by providing documents written in specific formats, using unambiguous language to send clearly assessable information. The reader in turn thoroughly understands the information in order to give a thoughtful response (Riordan, 2005).

Technical Writing: A Definition (Gerson, n.d.)

Technical writing is communication written for and about business and industry, focusing on products and services: how to manufacture them, market them, manage them, deliver them, and use them.

Technical writing is written in the work environment for supervisors, colleagues, subordinates, vendors, and customers

Technical writing, which must be understood easily and quickly, includes:

- memos and e-mail
- letters
- reports
- instructions
- brochures and newsletters
- the job search
- web pages
- fliers
- PowerPoint presentations
- Graphics

Technical writing is not literature; it's neither prose which recounts the fictional tales of characters nor poetry which expresses deeply felt, universal emotions through similes and metaphors.

Technical writing is neither an expressive essay narrating an occurrence nor an expository essay analyzing a topic.

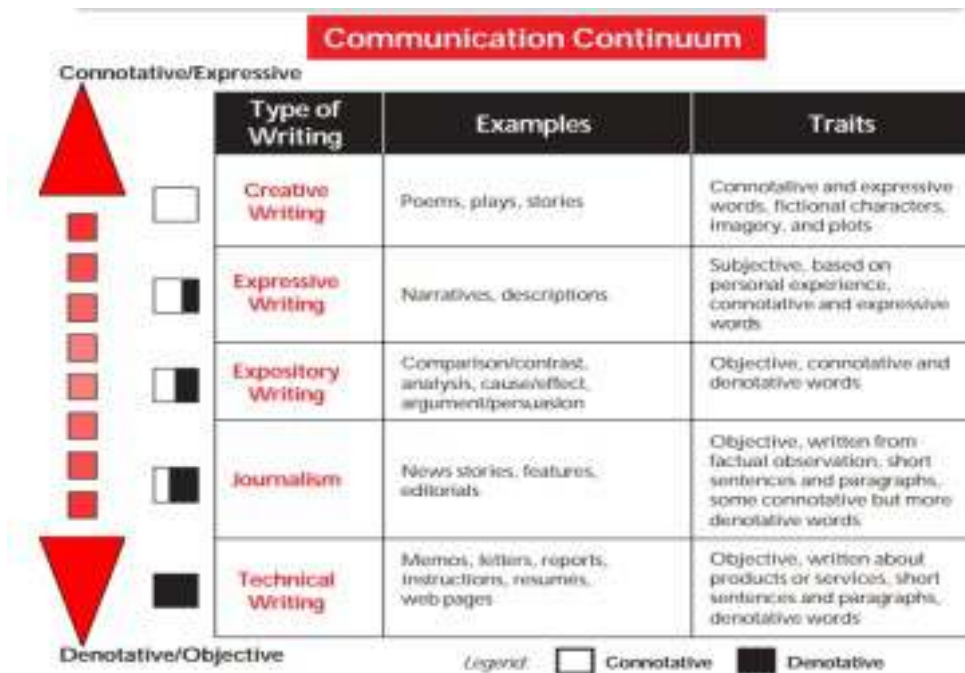
Technical writing is not journalism, written to report the news.

Technical writing does not focus on poetic images, describe personal experiences, or report who won the basketball game.

Instead, technical writing is:

- an instructional manual for repairing machinery
- a memo listing meeting agendas
- a letter from a vendor to a client

—a recommendation report proposing a new computer system
Study this communication continuum to further understand the difference of Technical writing from other types of writing.



Note: Of course, there are exceptions... Newsletters, sales letters, websites, and fliers might include promotional information. Such sales details could depend on expressive words—maybe even fictional characters. However, generally speaking, most technical writing is denotative versus connotative.

IMPORTANCE OF TECHNICAL COMMUNICATION

Technical communication plays a pivotal role in an organization, whether it is a business enterprise, an industry, or an academic institution. All managerial or administrative activities involve communication, be it planning, organizing, recruiting, coordinating, or decision-making. When you write reports, give instructions, or read brochures and manuals, you are involved in the process of communication. Communication serves as an instrument to measure the success or growth of an organization. For example, papers published by research and development organizations bring to light their progress.

- The success of any organization is largely recognized by the quality and quantity of information flowing through its personnel. When the CEO of an organization presents his company's achievements in a meeting, each of the participants comes to know of these milestones.
- Technical communication in an organization can be divided into two parts: oral and written. Both are equally important, as it cannot be specified which of these is used more in an organization.

It is not necessary that all these forms of communication should exist in an organization. If the organization is very large, for example, a university with

various affiliated institutions, or a business enterprise with branches across the nation and abroad, it may have all these forms of communication, whereas a smaller educational institution or business may not have all of them. Therefore, its importance can never be ignored.

THE QUALITIES OF GOOD TECHNICAL WRITING

Survey show that organizations rank writing skills in this order of importance: 1. Accuracy

2. Clarity
3. Conciseness
4. Readability
5. Usability
6. Correctness

These qualities mean that a document

- Makes a good impression when readers first interact with it: document is neat, readable, well organized, and inviting
- Can be read selectively – for instance, by some users, only the summary; by other users, only the introduction and conclusions; by still other users, the entire report
 - Shows a plan that reveals the purpose and value of the document
 - Makes sense: ideas appear in a logical sequence immediately evident from the document design
 - Uses visuals, if necessary to help readers understand ideas or data
 - Conveys an overall impression of authority, thoroughness, soundness, and honest work
 - Makes sense to people who were not part of the initial readership
 - Makes a positive statement about the writer and the organization
 - Enables people who need to use your writing to perform a task to do so. Beyond all these basic characteristics, good technical writing has no typographical errors, grammatical slips, and misspelled words. Little flaws distract attention from the writer's main points and call into question the writer's literacy.

COMMUNICATION CHANNELS (Gerson & Gerson, 2012)

Technical communication takes many different forms. Not only will you communicate both orally and in writing, but also you will rely on various types of correspondence and technology, dependent upon the audience, purpose, and situation. To communicate successfully in the workplace, you must adapt to many different channels of communication.

Table 1.1. gives you examples of different communication channels, both oral and written.

Many communication channels overlap in terms of purpose and audience. If you are requesting information from a vendor, for example, you could write a letter, send an e mail message, or make a telephone call. However, in other instances, communication

channels are more exclusive. You would not want to communicate bad news – such as layoffs, loss of benefits, or corporate closings – to employees by way of mass e-mail messages or televised reports. In these instances, face-to-face

meetings would be more appropriate. A key to successful technical communication is choosing the right channel.

TABLE 1.1 Communication Channels

Written Communication Channels	Oral Communication Channels
• E-mail • Memos • Letters • Reports • Proposals • Fliers • Brochures • Faxes • Web sites • Instant messages and text messages • Blogs • YouTube • Facebook • Twitter • Job information (resumes, letters of application, follow-up letters, interviews)	• Leading meetings • Conducting interviews • Making sales calls • Managing others • Participating in teleconferences and videoconferences • Facilitating training sessions • Participating in collaborative team projects • Providing customer service • Making telephone calls • Leaving voice mail messages • Making presentations at conferences or to civic organizations • Participating in interpersonal communication at work • Conducting performance reviews

MODULE 2

When the world ushered 21st century, many things changed and advanced in various fields and disciplines. These changes and advancements were prompted by the rapid pace of technology. Communication has been made easy and convenient.

Although this is the trend, we still can not do away with the traditional ways. In the field of language, the effective use of technical writing has been more pronounced most specially in the field of science, information technology and engineering.

Possessing the skills in technical writing is an advantage for those who are tasked to work on writing technical reports, manuals brochures and handbooks.

Direction : Study the differences between technical writing and literary writing.

	TECHNICAL WRITING	LITERARY WRITING
Content	Factual, straight forward	Imaginative, symbolic
Audience	Specific	General
Purpose	Inform, instruct, persuade	Entertain, provoke, captivate
Style	Formal, standard,	Informal, artistic, figurative

	academic	
Tone	Objective	Subjective
Vocabulary	Specialized	General, evocative
Organization	Sequential, systematic	Arbitrary, artistic

Principles of Technical Writing

Technical Writing is writing for a specific purpose and with a specific goal. Usually its goal is to inform/instructor persuade/argue. Technical writing can really be considered transactional writing because there are two people or groups involved in the communication. One party has a clear goal to inform or persuade the other party. This is real-world writing in every sense. That is why it is important to bear in mind the following:

1. Always have in mind a specific reader, real or imaginary, when you are writing a report. Always assume that he is intelligent but uninformed.
2. Before you start to write, always decide what the exact purpose of your report is, and make sure that every paragraph, every sentence, every word makes a clear contribution to that purpose.
3. Use language that is simple, concrete and familiar.
4. At the beginning and end of every section of report, check your writing according to this principle : "First tell the reader what you're going to tell him, then you tell him what you've told him".
5. Make your report attractive.

While bearing in mind the important principles of technical writing, it would also be helpful to distinguish the purpose of technical writing so that you would do not confuse with other writing forms.

3 | Page

- a. It serves as basis for management decision.
- b. It furnishes needed information.
- c. It gives instructions.
- d. It records business transactions through proposals.
- e. It procures business proposals.
- f. It serves as basis for public relations.
- g. It provides report to stockholders of companies.

Literary Texts vs Technical Texts

Technical writing and Literary writing are two important writing styles used by writers depending on the subject matter, purpose and intended audience. The main difference between technical writing and literary writing is that, literary

language is the writing style used in literary work while technical writing is a style used in writing for a particular field.

Literary writing is a style of writing that is used in creative and literary work; this is the style of writing that is used in fiction. Examples for literary writing includes poems, novels, short stories, dramas etc. The most significant difference between literary writing and other styles of writing is that the language used in literary writing uses many literary figures.

Technical writing is a style of writing used in delivering technical information regarding a particular subject. Here, the intended audience should have a certain knowledge about the subject in order to understand the technical jargon and the meaning of the text. Technical writing is the style of writing that is mostly observed in Non fiction. Examples for technical writing include essays, manuals, reports etc. This style of writing is direct and simple.

Literary writing	Technical writing
“I wandered lonely as a cloud That floats on high o’er vales and hills, When all at once I saw a crowd, A host, of golden daffodils; Beside the lake, beneath the trees, Fluttering and dancing in the breeze.” (First stanza from William Wordsworth’s “I wandered Lonely As a Cloud”)	The narrator was walking alone, when he saw a patch of daffodils near the lake.

Definition:

Technical writing: Technical writing is a process of managing technical information in a way that allow people to take actions.

Literary writing: Literary writing is a creating innovative, creative work, such as poems or novels, and compilations or volumes of creative work

Purpose:

Technical Writing: Written to inform, instruct readers about a certain thing. Literary Writing: Written to entertain, amuse readers.

Language

Technical Writing: The language used in technical writing is direct, factual, and straightforward.

Literary Writing: The language used in literary writing is creative, imaginative and uses literary techniques like hyperbole, personification, similes, metaphors, etc.

Appeal

Technical Writing: Technical Writing appeals to the mind.

Literary Writing: Literary Writing appeals to emotions.

Structure

Technical Writing: Technical writing has technical vocabulary, simple sentences, impersonal, objective tone.

Literary Writing: Literary writing might have complex sentence structure and linguistic aspects like dialects, ambiguity, etc.

Audience

Technical Writing: Technical writing is written for those who are knowledgeable about that particular subject area.

Literary Writing: Literary writing is written for general readers.

Dominant Patterns in Technical Writing

Definition

Language can be a source of confusion in the workplace as well. When technical terms are not understood or interpreted correctly, readers may not know what the document is saying or what is expected of them. That's why we need to include definitions as part of our technical writing. Definitions in technical writing refer to explaining the meaning of a word as it relates to the subject being discussed. Definitions can solve problems and clear up confusion within a document. Providing a definition for your readers will enable the writer and the audience to have a unified understanding of a term. Writing definition is used when :

- a. describing and defining new products and services
- b. introducing new information to the reader in clear and effective ways
- c. allowing experts and specialists to communicate with a more generalized audience

Methods of Defining Terms

Simple definition could be formulated by bearing in mind its three parts : the species, genus and differentia. This method of definition is commonly used in technical writing whether formally, semi-formally or non-formally.

Species (term to be defined)	Genus (the class where the term belongs)	Differentia (a characteristic of the term)
Language	is a form	of communication.
An architect	is a professional	who designs buildings.
Communication	Is a process	of giving and receiving information.

Another method of defining a term is by expanded definition. This is done by stipulation, operation, explication, cause and effect, classification, example and other rhetorical functions.

species

Communication is a
genus
systematic process in which people
interact with and through symbols to

create and interpret meanings.

differentia

Informal definition on the other hand does not follow a pattern. The only objective is to give meaning to a word that is unfamiliar or explain the special meaning of a familiar word. Word meanings enter informal definition by means of providing denotative or connotative meaning.

Description of a Process

There are writing tasks that entail an explanation of how something works, how something is done or how something is made. That is why knowing how to describe a process will enable a technical writer and his/her reader to perform or do something.

In describing a process, a writer explains the arrangement of a sequence in chronological order. As it is used in technical writing, the process is similar to mechanism description. Hence, process description includes sequence, instructions and procedure.

Description of Mechanism

Description of a mechanism or apparatuses is an explanation of a system or parts of an apparatus. This includes the characteristics and functions of a piece of the device and the totality of the mechanism.

It also explains the arrangement and shape of an object in space. In writing a description of a mechanism, the writer may be guided by the following questions:

- What is it? (equipment, machine, device, apparatus)
 - What is its function?
 - What does it look like?
 - How does it work?
 - What are its principal parts?

Classification

Classification is dividing something into groups, classes, categories, etc. This is normally done in accordance with several criteria (standards or principles on which judgements are based).

How to classify

One way of classifying objects or ideas is to look for relationships among them and to organize them into groups. It may either be on the basis of their similarities and differences.

Try to find the similarities or differences of the words listed below by classifying them into identifiable categories.

Audience Analysis

The audience of a technical report—or any piece of writing for that matter—is the intended or potential reader or readers. For most technical writers, this is the most important consideration in planning, writing, and reviewing a document. You “adapt” your writing to meet the needs, interests, and background of the readers who will be reading your writing. In reality, the lack of audience analysis and adaptation is one of the root causes of most of the problems you find in professional, technical documents— particularly instructions where it surfaces most glaringly.

Types of audiences

One of the first things to do when you analyze and audience is to identify its type (or types—it’s rarely just one type). The common division of audiences into categories is as follows:

- a. Experts: These are the people who know the business or organization (and possibly the theory and the product) inside and out. They designed it, they tested it, they know everything about it. Often, they have advanced degrees and operate in academic settings or in research and development areas of the government and technology worlds.
- b. Technicians: These are the people who build, operate, maintain, and repair the items that the experts design and theorize about. Theirs is a highly technical knowledge as well, but of a more practical nature.
- c. Executives: These are the people who make business, economic, administrative, legal, governmental, political decisions about the products of the experts and technicians. Executives are likely to have as little technical knowledge about the subject as non-specialists. For many of you, this will be the primary audience for your report.

- d. Non specialists: These readers have the least technical knowledge of all. They want to use the new product to accomplish their tasks; they want to understand the new power technology enough to know whether to vote for or against it in the upcoming bond election. Or, they may just daffodils. about a specific technical matter and want to learn about it—but for no specific, practical reason. Chances are these readers will represent your secondary audience.

Synthesis

Synthesizing simply means combining. Instead of summarizing the main points of each source in turn, you put together the ideas and findings of multiple sources in order to make an overall point. At the most basic level, this involves looking for similarities and differences between your sources.

In writing, synthesis is used in a similar way by supporting an idea or argument by showing relationships among a variety of sources. While there are many ways of employing synthesis in your writing, one method you can use is the OCIE Method; or, Observe, Cite, Interpret, and Evaluate or Exit.

MODULE 3

Technical documentation is intended to communicate information to the people who need it in a way that is clear and easy to read, at the right time to help make decisions and to support productivity. Designing technical communication is like designing any other product for an intended user: the ultimate goal is to make it “user friendly.” (Last, 2019)

LANGUAGE PROFICIENCY (Last, 2019)

Two key characteristics of professional technical communication are that it is precise and concise. This precision and concision must be evident at all levels, from the overall document, to paragraphing, to sentence structure to word choice, and even to punctuation. Every word or phrase should have a distinct and useful purpose. If it doesn't, cut it or revise.

The 7 Cs of Professional Writing

The 7 C's are simply seven words that begin with C that characterize strong professional style. Applying the 7 C's of professional communication will result in writing that is

- Clear
- Coherent
- Concise
- Concrete
- Correct
- Complete
- Courteous.

CLEAR writing involves knowing what you want to say before you say it because often a lack of clarity comes from unclear thinking or poor planning; this,

unfortunately, leads to confused or annoyed readers. Clear writing conveys the purpose of the document immediately to the reader; it matches vocabulary to the audience, avoiding jargon and unnecessary technical or obscure language while at the same time being precise. In clarifying your ideas, ensure that each sentence conveys one idea, and that each paragraph thoroughly develops one unified concept.

COHERENT writing ensures that the reader can easily follow your ideas and your train of thought. One idea should lead logically into the next through the use of transitional words and phrases, structural markers, planned repetition, sentences with clear subjects, headings that are clear, and effective and parallel lists. Writing that lacks coherence often sounds “choppy” and ideas seem disconnected or incomplete. Coherently connecting ideas is like building bridges between islands of thought so the reader can easily move from one idea to the next.

CONCISE writing uses the least words possible to convey the most meaning while still maintaining clarity. Avoid unnecessary padding, awkward phrasing, overuse of “to be” forms (is, are, was, were, am, be, being), long preposition strings, vagueness, unnecessary repetition and redundancy. Use active verbs whenever possible and take the time to choose a single word rather than a long phrase or clichéd expression. Think of your word count like a budget; be cost effective by making sure every word you choose does effective work for you. Cut a word, save a buck! As William Zinsser asserts, “the secret of good writing is to strip every sentence to its cleanest components.”

CONCRETE writing involves using specific, precise language to paint a picture for your readers so that they can more easily understand your ideas. If you have to explain an abstract concept or idea, try to use examples, analogies, and precise language to illustrate it. Use measurable descriptors whenever possible; avoid vague terms like “big” or “good.” Try to get your readers to “see” your ideas by using specific terms and descriptions.

CORRECT writing uses standard English punctuation, sentence structure, usage, and grammar. Being correct also means providing accurate information, as well as using the right document type and form for the task.

COMPLETE writing includes all requested information and answers all relevant questions. The more concrete and specific you are, the more likely your document will be complete as well. Review your checklist of specifications before submitting your document to its intended reader.

COURTEOUS writing entails designing a reader-friendly, easy-to-read document; using tactful language and appropriate modes of addressing the audience; and avoiding potentially offensive terminology, usage, and tone. As we have discussed in an early section, without courtesy you cannot be constructive.

In some cases, some of these might come into conflict: what if being too concise results in a tone that sounds terse, or an idea that seems incomplete? Figure 1 illustrates one method of putting all the 7Cs together.



SENTENCE STRUCTURES

A. SENTENCE VARIETY AND LENGTH

While variety makes for interesting writing, too much of it can also reduce clarity and precision. Technical writing tends to use simple sentence structures more often than the other types. That said, simple does not necessarily mean “simplistic,” short, or lacking in density. Remember that in grammatical terms, simple just means that it has one main clause (one subject and one predicate). You can still convey quite a bit of concrete information in a simple sentence.

The other consideration for precise writing is length. Your sentences should vary in length just as they can vary in type. However, you want to avoid having too many long sentences because they take longer to read and are often more complex. That is appropriate in academic writing but less so in technical writing. The goal is to aim for an average of around 20 to 30 words per sentence. Reserve the short sentences for main points and use longer sentences for supporting points that clarify or explain cause and effect relationships. If you feel the sentence is too long, break it into two sentences. You do not want your reader to have to read a sentence twice to understand it. If you make compound or complex sentences, ensure that you use appropriate coordinating or subordinating strategies to make the relationship between clauses perfectly clear. See Appendix E to review specific information on simple, compound, and complex sentence structures.

B. PRECISE WORDING

Technical writing is precise writing. Vague, overly general, hyperbolic or subjective/ambiguous terms are simply not appropriate in this genre. You do not want to choose words and phrasing that could be interpreted in more than one way. For example, if you asked someone to define what makes a “good dog,”

you might get responses like “obedient, effective hunter/retriever, well-behaved, affectionate, loyal, therapeutic, goofy” and “all dogs are good!” Choose words that most precisely, concisely, and accurately convey the idea you want to convey. Below are some guidelines and examples to follow for using precise wording.

1. Replace abstract nouns with verbs.

Verbs, more than nouns, help convey ideas concisely, so where possible, avoid using nouns derived from verbs. Often these abstract nouns end in -tion and -ment. See examples in the following chart.

Abstract Noun	Verb
Acquisition	acquire
analysis	analyze
recommendation	recommend
observation	observe
application	apply
confirmation	confirm
development	develop
ability	able, can
assessment	assess

2. Prefer short words to long words and phrases.

The goal is to communicate directly and plainly so use short, direct words whenever possible. In other words, don't use long words or phrases when short ones will do. Write to express, not impress

Long	Short
cognizant; be cognizant of	aware, know
commence; commencement	begin, beginning
utilize; utilization	use (v), use (n)
inquire; make an inquiry	ask
finalize; finalization	complete, end
afford an opportunity to	permit, allow
at this point in time	now, currently
due to the fact that	because, due to
has the ability to	can

3. Avoid clichés.

Clichés are expressions that you have probably heard and used hundreds of times. They are over-used expressions that have largely lost their meaning and impact.

Clichés	Alternatives
as plain as day	plainly, obvious, clear
ballpark figure	about, approximately
few and far between	rare, infrequent
needless to say	of course, obviously

last but not least finally, lastly
as far as ___ is concerned ?

4. Avoid cluttered constructions.

This category includes redundancies, repetitions, and “there is/are” and “it is” constructions.

Redundancies		
combine/join together	fill completely	unite as one
finish entirely	refer/return/revert back to	emphasize/stress strongly
examine (closely)	suddenly interrupt	better/further enhance
eventually evolve over time	strictly forbid	rely/depend heavily
plan ahead	harshly condemn	protest against
completely surround on all sides	estimate/approximate roughly	gather/assemble together
clearly articulate	carefully consider	successfully prove
future plan	mutual agreement	years of age
in actual fact	positive benefits	end result/product

5. Use accurate wording.

Sometimes this requires more words instead of fewer, so do not sacrifice clarity for concision. Make sure your words convey the meaning you intend. Avoid using words that have several possible meanings; do not leave room for ambiguity or alternate interpretations of your ideas. Keep in mind that readers of technical writing tend to choose literal meanings, so avoid figurative language that might be confusing (for example, using the word “decent” to describe something you like or think is good). Separate facts from opinions by using phrases like “we recommend,” “we believe,” or “in our opinion.” Use consistent terminology rather than looking for synonyms that may be less precise.

Qualify statements that need qualifying, especially if there is possibility for misinterpretation. Do not overstate through the use of absolutes and intensifiers. Avoid overusing intensifiers like “extremely,” and avoid absolutes like “never, always, all, none” as these are almost never accurate. Remember Obiwan Kenobi’s warning:

“Only a Sith deals in absolutes.”

We tend to overuse qualifiers and intensifiers, so below are some that you should be aware of and consider whether you are using them effectively.

Overused Intensifiers

absolutely actually assuredly certainly clearly completely
considerably definitely effectively extremely fundamentally
drastically
highly in fact incredibly inevitably indeed interestingly
markedly naturally of course particularly significantly surely totally
utterly very really remarkably tremendously

Overused Qualifiers

apparently arguably basically essentially generally hopefully in effect
in general kind of overall perhaps quite
rather relatively seemingly somewhat sort of virtually

READER'S LEVEL OF KNOWLEDGE

Writing can be conceptualized as writer-centred or reader-centred. Things like diaries and journals are primarily writer-centred, in that they are written for the benefit of the writer. Your schoolwork may also have been somewhat writer-centred, in that often your goal was to “show what you know” and thereby “get a good grade.” Technical communications require that you shift this mindset and write for the benefit of your reader—or design the content and structure of your communication for your “user.” This mindset should be informed by an understanding of your audience. Use these guidelines and ask yourself the following questions:

- **Who is my target audience?** Are they internal or external readers? Upstream, downstream or lateral from you? Do I have multiple readers?
- **What is their perspectives on the topic, on me, and on the document I will write?** What are they expecting to do with the document? What is the document meant to accomplish? Why has it been requested? What is my role and relationship to my readers? What does the reader need to know? Already know? What does my reader NOT need to have explained?
- **What is my goal or purpose in writing to these readers?** What am I trying to communicate? What do I want them to do as a result of reading this document? How can I plan the content to meet my readers' needs?
- **What is my reader's goal?** Why does this audience want or need to read this document?

Getting a clear understanding of your audience is important in communicating effectively. It also enables you to imagine your audience as you write and revise. Keep asking yourself whether what you have said would be clear to your audience. How could you say it better?

Types of audiences (McMurrey, 2017)

One of the first things to do when you analyze an audience is to identify its type (or types—it's rarely just one type). The common division of audiences into categories is as follows:

- **Experts:** These are the people who know the business or organization (and possibly the theory and the product) inside and out. They designed it, they tested it, they know everything about it. Often, they have advanced degrees and operate in academic settings or in research and development areas of the government and technology worlds.
- **Technicians:** These are the people who build, operate, maintain, and repair the items that the experts design and theorize about. They have a highly technical knowledge as well, but of a more practical nature.
- **Executives:** These are the people who make business, economic, administrative, legal, governmental, political decisions about the products of the experts and technicians. Executives are likely to have as little technical knowledge about the subject as nonspecialists. For many of you, this will be the primary audience for your report.
- **Nonspecialists:** These readers have the least technical knowledge of all. They want to use the new product to accomplish their tasks; they want to understand

the new power technology enough to know whether to vote for or against it in the upcoming bond election. Or, they may just be curious about a specific technical matter and want to learn about it—but for no specific, practical reason. Chances are, these readers will represent your secondary audience.

Audience analysis

It's important to determine which of the four categories just discussed represent your potential audience(s), but that's not the end of it. Audiences, regardless of category, must also be analyzed in terms of characteristics such as the following:

- **Background—knowledge, experience, training:** One of your most important concerns is just how much knowledge, experience, or training you can expect in your readers. If you expect some of your readers to lack certain background, do you automatically supply it in your document? Consider an example: imagine you are writing a guide to using a software product that runs under Microsoft Windows. How much can you expect your readers to know about Windows? If some are likely to know little about Windows, should you provide that information? If you say no, then you run the risk of customers getting frustrated with your product. If you say yes to adding background information on Windows, you increase your work effort and add to the page count of the document (and thus to the cost). Obviously, there's no easy answer to this question—part of the answer may involve just how small a segment of the audience needs that background information.
- **Needs and interests:** To plan your document, you need to know what your audience is going to expect from that document. Imagine how readers will want to use your document; what they will demand from it. For example, imagine you are writing a manual on how to use a new smartphone—what are your readers going to expect to find in it? Imagine you are under contract to write a background report on global warming for a national real estate

association— what do readers want to read about; and, equally important, what do they not want to read about?

- **Other demographic characteristics:** And of course there are many other characteristics about your readers that might have an influence on how you should design and write your document—for example, age groups, type of residence, area of residence, gender, political preferences, and so on.

Audience analysis can get complicated by at least two other factors: mixed audience types for one document, wide variability within audience, and unknown audiences.

- **More than one audience:** You are likely to find that your report is for more than one audience. For example, it may be seen by technical people (experts and technicians) and administrative people (executives). What should you do in this case? You can either write all the sections so that all the audiences of your document can understand them. Or you can write each section strictly for the audience that would be interested in it, then use headings and section introductions to alert your audience about where to find relevant information in your report.
- **Wide variability in an audience:** You may realize that, although you have an audience that fits into only one category, its background varies widely. This is a tough one—if you write to the lowest common denominator of reader, you are likely to end up with a cumbersome, tedious book-like report that will turn off the majority of readers. However, if you don't write to that lowest level, you lose that segment of your readers. What should you do? Most writers go for the majority of readers and sacrifice that minority that needs more help. Others put the supplemental information in appendixes or insert cross-references to beginners' books.

Adapting your writing to meet your audience's needs

Once you've analyzed your audience, how do you use this information? How do you keep from writing something that may potentially still be incomprehensible or useless to your readers? Draft your document with your audience's needs in mind, but remember that writing can be refined over many drafts. With each subsequent draft, think more carefully about your readers, and revise and edit your document so that you make technical information more understandable for nonspecialist audiences. The lists below are some of the ways you can adapt your writing to your audience's needs.

The following "controls" have mostly to do with making technical information more understandable for nonspecialist audiences and is information you will refine as you begin to put your final report together. However, it is a good idea to be aware of your audience's needs even in the early stages of your report drafting.

Provide the right information

Add information readers need to understand your document. Check to see whether certain key information is missing—for example, a critical series of

steps from a set of instructions; important background that helps beginners understand the main discussion; definition of key terms.

Omit information your readers do not need. Unnecessary information can also confuse and frustrate readers—after all, it’s there so they feel obligated to read it. For example, you can probably chop theoretical discussion from basic instructions.

Change the level of the information you currently have. You may have the right information but it may be “pitched” at too high or too low a technical level. It may be pitched at the wrong kind of audience—for example, at an expert audience rather than a technician audience. This happens most often when product-design notes are passed off as instructions.

Add examples to help readers understand. Examples are one of the most powerful ways to connect with audiences, particularly in instructions. Even in a non-instructional text, for example, when you are trying to explain a technical concept, examples are a major help—analogy in particular.

Change the level of your examples. You may be using examples but the technical content or level may not be appropriate to your readers. Homespun examples may not be useful to experts; highly technical ones may totally miss your nonspecialist readers.